Materials, stress & strain

Higher · Mechanisms & structures · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate the strain.

- $\varepsilon = \Delta L \div L = 1.2 \div 2000$
- $\varepsilon = 6.0 \times 10^{-4}$ (no units)

No marks for: Mixing metres and millimetres

3 Quick check

- a) It is corrosion resistant, so it won't degrade in sea water
- b) To cancel temperature effects
- c) Brittle
- d) 143 N/mm²

4 Key terms

- 1 – D
- 2 – C
- 3 – A
- 4 – E
- 5 – F
- 6 – B

5 Choose a suitable material from aluminium, mild steel and carbon fibre, and justify your choice.

- Carbon fibre (or aluminium)
- Justification linked to property: high strength-to-weight ratio, does not corrode in seawater

No marks for: Mild steel: it corrodes and is heavy

6 Calculate the factor of safety.

- FoS = UTS ÷ working stress = 430 ÷ 85
- FoS = 5.1

7 Give two reasons for the advice.

- Loads could be higher than expected (crowds, wind)
- Material may have defects or weaken through corrosion/fatigue
- Failure would endanger lives